

Conglomerate Mergers: Effects on Growth and Competition

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Abstract

Your abstract goes here. It should be a brief summary of your article.

1 Introduction

[2]

2 Background

Compound annual growth rate volatility tax, variance drain [1], [5] [3] Merge as a hedge against larger draw downs

$$g = \mu - \frac{\sigma^2}{2} \quad (1)$$

3 Model

3.1 Random Multiplicative Growth

3.1.1 Growth Process

To investigate conglomerate formation we set up an Agent Based Model with M markets that are uncorrelated between each other, each containing an amount of N firms. Firms face random multiplicative growth that is unique for their market, drawn from a normal distribution with positive mean $\mathcal{N}(\mu_m, \sigma_m)$. In this framework each firms state is multiplicatively linked to its previous one. This can be interpreted as firms setting a certain price and obtaining a random demand for it.

$$s_{it+1} = s_{it}\delta_t \quad \text{with} \quad \delta_t \sim \mathcal{N}(\mu_m, \sigma_m) \quad (2)$$

We model markets such that mean and the standard deviation of the distribution over their growth increments are positively correlated, meaning that a higher mean μ_m comes with a higher standard deviation σ_m . As explained in section 2, in multiplicative processes one is not just maximizing for the ensemble mean, but for the growth rate. Hence, a higher scale σ_m decreases the growth rate, through the volatility tax.

Each firm thus realizes a gain of $\Delta_{it} = s_{it}\delta_t - s_{it-1}$. For any firm, not part of a conglomerate this will also be their period profit $\Pi_{it} = \Delta_{it}$. Firms in conglomerates pool an amount $\alpha\Delta_{it}$, which flows into the conglomerate's period pool $\Omega_{ct} = \sum_{i \in c} \alpha\Delta_{it}$. However, as conglomerates grow in size, maintaining them becomes ever more difficult as transaction costs increase. We model these as a logistic function of the form

$$\Phi(O_c, M) = \frac{1}{1 + \exp^{-|O_c| + 0.4M}} \quad (3)$$

This factor scales the cost according to conglomerate size $\sum_{i \in c} s_{it}$ such that the pool size is given by

$$\Omega_{ct} = \sum_{i \in c} \alpha \Delta_{it} - \Phi(O_c, M) \sum_{i \in c} s_{it} \quad (4)$$

Notice that the factor 0.4 in logistic function scales, the costs so that a conglomerate operating in 40 % of all markets would lose 50 % of their market value in transaction cost every round. Lastly, this pool will be shared equally such that a members profit is determined by $\Pi_{it} = (1 - \alpha)\Delta_{it} + \frac{\Omega_{ct}}{|O_{c_i}|}$. Consequently, any firms next state is given by $s_{it+1} = s_{it} + \Pi_{it}$.

3.1.2 Merging Process

At the beginning of every time step t each firm $i \in M \times N$ gets a chance to choose another with probability p . If successful, it will select a random other firm j , among the remaining $(M \times N) - 1$ firms. We allow conglomerates C to only operate one firm per market, meaning it can have a maximum of M members, with the set markets of a conglomerate defined as $O_c \subseteq M$. Hence, conglomeration is only *feasible* if firm j and i :

1. don't operate in the same market ($m_i \neq m_j$)
2. are not part of a conglomerate that operates in a market in which the other firm, or the conglomerate which it is part of operates ($O_{c_i} \cap O_{c_j} = \emptyset$)

We assume that if both firms are in conglomerates that these will merge as well. After confirming *feasibility*, merging firms determine desirability, meaning that they would both benefit from joining forces. They do so by comparing the realized pool Ω size over the last l periods, against a "synthetic" pool $\hat{\Omega}$ size if they had merged. If $\hat{\Omega} > \Omega$ the merger will be implemented.

3.1.3 Splitting Process

After an initial l periods each firm that is part of a conglomerate evaluates its performance within it by comparing its growth rate $g_{i(t-l):t}$ by being part of the conglomerate against the potential rate $\hat{g}_{i(t-l):t}$ it had realized, had it remained outside. Upon being worse off, a firm exits the conglomerate and continues to operate alone.

3.2 Endogenous Pricing

Going beyond random growth, we enable our firms to optimize revenues by setting their period prices p_{it} for which they receive the following demand

$$D_{it}(\mathbf{P}_t, \theta_{it}) = 1 - (1 - \theta_t)p_{it} + 0.5 \max_{j:j \neq i} \mathbf{P}_t \quad (5)$$

where θ_{it} is the firms current market share, which are the relative cumulative profits within a market

$$\theta_{it} = \frac{\sum_{\tau=1}^t \Pi_{i\tau}}{\sum_{\tau=1}^t \sum_{j:j \neq i} \Pi_{j\tau}} \quad (6)$$

Consequently, period revenues take the form of $D_{it}(\mathbf{P}_t, \theta_{it})p_{it}$. We further assume stochastic costs γ that are drawn from a log-normal distribution $\mathcal{L}(0, 1)$ that is scaled to match the mean growth rate of revenues per period. Hence, a firm's pre-pooling profit is given by $\Delta_{it} = D_{it}(\mathbf{P}_t, \theta_{it})p_{it} - \gamma$. If the firm is not in a conglomerate, there is no pooling, meaning $\Pi_{it} = \Delta_{it}$. In case it is part of a conglomerate, it will continue as outlined in section 3.1.

To learn the optimal pricing policy π , we trained a reinforcement learning agent using the Proximal Policy Optimization algorithm [4]. We trained the agents in a separate environment being tasked to only optimize revenues $D_{it}(\mathbf{P}_t, \theta_{it})p_{it}$. Its information set contains its price and market share in the previous round, as well as the maximum price of its competitors in the previous two rounds, $\{p_{it-1}, \theta_{it-1}, \max_{j:j \neq i} \mathbf{P}_{jt-1}, \max_{j:j \neq i} \mathbf{P}_{jt-2}\}$. To expose the agents to possible market share constellations, we exogenously set the θ_{it} . The resulting policy can be seen in figure 1.

The resulting model flow can be seen in algorithm 1.

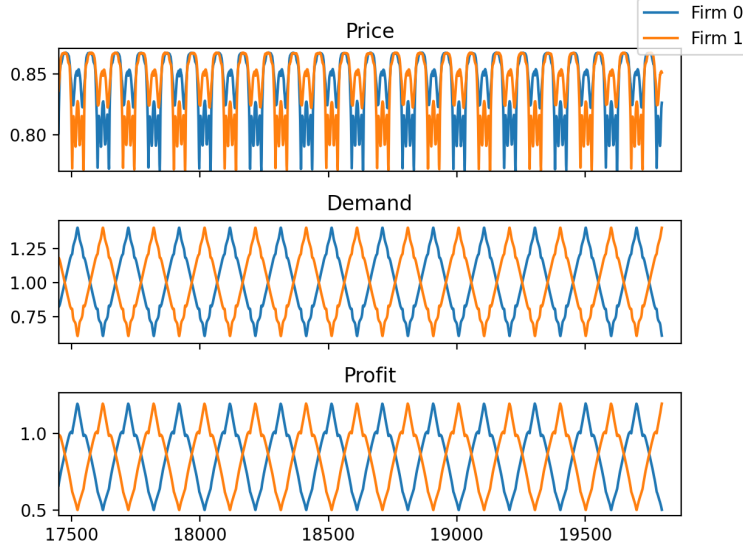


Figure 1: Testing a trained policy with countervailing market shares.

Algorithm 1: Model flow with endogenous pricing.

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for  $t$  in  $T$  do
  foreach  $i \in N \times M$  do
     $\lfloor$  Merge with  $j$  if  $(m_i \neq m_j)$  and  $(O_{c_i} \cap O_{c_j} = \emptyset)$  and  $\hat{\Omega} > \Omega$ 
  foreach  $i \in N \times M$  do
     $\lfloor$   $p_{it} = \pi(p_{it-1}, \theta_{it-1}, \max_{j:j \neq i} \mathbf{P}_{jt-1}, \max_{j:j \neq i} \mathbf{P}_{jt-2})$ 
   $\Delta_t = D(\mathbf{P}_t, \theta_t) \mathbf{P}_t - \gamma$ 
  foreach  $c \in C$  do
     $\Omega_{ct} = \sum_{i \in c} \alpha \Delta_{it} - \Phi(O_c, M) \sum_{i \in c} s_{it}$ 
    foreach  $i \in c$  do
       $\lfloor$   $\Pi_{it} = (1 - \alpha) \Delta_{it} + \frac{\Omega_{ct}}{|O_c|}$ 
  foreach  $i \notin c$  for all  $c \in C$  do
     $\lfloor$   $\Pi_{it} = \Delta_{it}$ 
  foreach  $i \in M \times N$  do
     $\lfloor$   $\theta_{it} = \frac{\sum_{\tau=1}^t \Pi_{i\tau}}{\sum_{\tau=1}^t \sum_{j:j \neq i} \Pi_{j\tau}}$ 
  foreach  $i \in c$  for all  $c \in C$  do
     $\lfloor$  Exit  $c$  if  $\hat{g}_{i(t-l):t} > g_{i(t-l):t}$ 

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- 4 Results
- 5 Discussion
- 6 Conclusion

References

- [1] Tom Messmore. Variance drain. *Journal of Portfolio Management*, 21(4):104, 1995.
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- [4] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- [5] Mark Spitznagel. *Safe Haven: Investing for Financial Storms*. John Wiley & Sons, 2021.